

Statistically Controlling for Processing Fluency Reduces the Aesthetic-Usability Effect

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ABSTRACT

The aesthetic-usability effect asserts that user interfaces that appear aesthetic also appear easier to use. Most explanations for the effect see aesthetics in a causal role. In contrast, we propose that processing fluency as a third variable causes both judgements of aesthetics and usability. Processing fluency refers to the subjective ease of information processing and has been shown to influence, among others, judgements of aesthetics and usability. We tested our proposition in an experiment in which users rated screenshots of city websites. The aesthetic-usability effect was replicated by our data: aesthetics and usability correlated .79. When controlling for fluency, however, the aesthetic-usability effect was considerably diminished; the correlation decreased to .34. Future research will address the limitations of this study by investigating a wider range of website designs and adding interactivity to the interfaces.

CCS CONCEPTS

• **Human-centered computing** → Human computer interaction (HCI); HCI theory, concepts and models; Human computer interaction (HCI); Empirical studies in HCI.

KEYWORDS

aesthetic-usability effect, processing fluency, expected usability, websites

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1 INTRODUCTION

"What is beautiful is usable": more than 20 years ago, Tractinsky et al. [49] showed that users judge aesthetic user interfaces to be more usable than unaesthetic interfaces. This relationship between perceived aesthetics and perceived usability, termed the aesthetic-usability effect, has been replicated several times since (Table 1). Most of these studies report a positive correlation between perceived aesthetics and perceived usability of a user interface. Some studies also experimentally manipulated aesthetics and showed

that aesthetic user interfaces are not only perceived to be easier to use, but also that aesthetics can affect objective performance measures [46]. Other studies, however, found a reverse influence of usability on aesthetics [13, 51].

Several explanations for these relationships have been offered. First, there could be a stereotypical association that aesthetic websites are also easy to use [49]. Second, there might be a halo effect, in which the aesthetic impression radiates to other properties such as usability and thus influences the perception thereof [49]. Third, aesthetic design can have a positive effect on the mood of the users, and potential usability problems are thus perceived to be easier to overcome [31]. Fourth, the visual clarity of a user interface is a common component of aesthetics and usability measures and has been found to mediate the effect of perceived aesthetics on perceived usability [26, 38]. Common to these explanations is the idea that aesthetics play a causal role, and its effect on usability is mediated by other variables. We would like to propose a different explanation: that there is a hidden third variable that influences both aesthetics and usability ratings, causing their interrelation. We propose that this third variable is processing fluency, a concept mainly investigated in cognitive social psychology.

Processing fluency describes the subjective ease with which information is perceived and processed in the mind [12]. Psychological studies show that processing fluency affects various judgement domains. Fluent stimuli tend to be liked more than disfluent stimuli; they appear to be more familiar, less risky; they are trusted more and are perceived as more aesthetically pleasing and attractive [1, 12]. Although processing fluency has become a staple of research in cognitive and social psychology, it has received little attention in HCI research. The aim of this paper is to transfer the findings of psychological research to HCI and to investigate the claim that processing fluency, as a hidden third variable, could explain the aesthetic-usability effect. Based on a pre-study, fluent and disfluent websites were selected as stimuli. In the main study, users rated them in terms of perceived aesthetics, perceived usability, and processing fluency. Thus, this study contributes (1) a replication of the aesthetic-usability effect and (2), by manipulating and controlling for processing fluency, an investigation of how much of this effect can be attributed to processing fluency as an influencing third factor.

In the following, we introduce previous research on processing fluency and the aesthetic-usability effect. Then, we report our study and discuss the results. This is followed by a critical reflection and an outlook for future research.

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Table 1: Overview of relevant studies on the relationship between aesthetics and expected usability

Researcher	Product (task)	<i>r</i>	Aesthetics measure	Usability measure
Kurosu & Kashimura [22]	ATM layouts (viewing)	.59	self-designed (1 item)	self-designed (1 item)
Tractinsky [48]	ATM layouts (viewing)	.83 – .92	Kurosu & Kashimura [22]	Kurosu & Kashimura [22]
Tractinsky et al. [49]	ATM layouts (viewing)	.66	Kurosu & Kashimura [22]	Kurosu & Kashimura [22]
Sonderegger & Sauer [43]	Smartphones (viewing)	.19	self-designed (1 item)	self-designed (1 item)
Papachristos & Avouris [34]	Websites (viewing)	.45	self-designed (1 item)	self-designed (1 item)
Tuch et al. [50]	Websites (viewing)	.33	Hassenzahl [14] (1 item)	self-designed (1 item)
Schrepp & Müller [38]	Websites (viewing)	.44	self-designed (1 item)	self-designed (1 item)

Source: Adapted from [15, 47]

2 RELATED WORK

2.1 Processing Fluency

The concept of processing fluency refers to the subjective ease with which we process information [41]. Psychological research found that this subjective feeling accompanies each cognitive activity and acts as a metacognitive cue, determining the type and content of information processing [1]. Processing fluency can have different sources, notably the *perceptual* and *conceptual* ease of processing [35]. Perceptual fluency, thus, refers to the ease with which the physical properties of a stimulus are perceived. In HCI, it can be influenced by the design characteristics of a user interface, e.g., its shape, size, figure-ground contrast, and visual complexity [35]. Perceptual fluency is also influenced by the repeated exposure or the duration of stimulus presentation [11]. Conceptual fluency refers to the ease of identifying the meaning of a stimulus and can be facilitated by conceptual priming [56]. Conceptual priming means that the activation of mental representations in one context unconsciously affects representations in a subsequent and independent context [4]. Depending on the cognitive task involved, Alter et al. [1] distinguish other forms of fluency, including linguistic fluency, memory retrieval fluency, embodied cognitive fluency, decision fluency, and prospective imagery fluency. But regardless of its origin, the feeling of fluency is always a similar phenomenological experience [42]. Hence, regardless of its source, there is only one relevant type of processing fluency that affects further information processing.

Processing fluency has been shown to influence several domains of judgement. Processing fluency can determine how much a stimulus is perceived as being aesthetic [35] and how much it is liked [36]. Similarly, fluent product designs are judged to be more aesthetic [24] and are liked more [23, 29]. The effect of fluency on judgement domains is called the fluency effect [27]. Other domains with fluency effects include estimating how true a statement is [7] or how risky it is to use a product [44] (see [1, 12, 42] for comprehensive overviews).

Two explanations account for fluency effects. First, according to the fluency attribution hypothesis [32], the feeling of fluency is used as a source of metacognitive information. Thus, fluency is misinterpreted as factual information about the stimulus at hand. For example, although the feeling of fluency could origin from a good figure-ground contrast, low visual complexity of a website and a legible font, users could misattribute this feeling of fluency

to being familiar with the website and thus judge it as easier to use. Second, according to the hedonic marking hypothesis [56], high fluency elicits a positive emotional response that has a particular impact on affective judgements such as liking and aesthetics [54]. Note that it is the subjective feeling of fluency that produces fluency effects on other judgements. It likely is influenced by – but differs from – objective fluency that could be measured as mental workload [12].

2.2 Aesthetic-Usability Effect

As early as 1995, Kurosu and Kashimura [21] investigated perceived usability in relation to perceived aesthetics. In their study, participants judged images of ATM layouts. The results showed that measures of perceived aesthetics correlated strongly with measures of perceived ease of use. Subsequently, several studies investigated the relationship between aesthetics and expected usability (Table 1). Tractinsky [48] found a strong relationship between aesthetics and usability with a student sample. Tractinsky et al. [49] showed that the correlation continued to increase after usage. They summarised their results by stating, “What is beautiful is usable,” which Lidwell et al. [28] took up and named the aesthetic-usability effect. In addition to correlative studies, experimental studies manipulated either aesthetics or usability and examined how changes in the UI design affected the perceived usability or the perceived aesthetics, respectively. Some studies found the proposed effect of aesthetics on usability, and some also found a reversed effect of usability on aesthetics (“What is usable is beautiful” [13, 50]).

Tuch et al. [51] provide a review of correlative and experimental studies. Most studies measure usability using single questionnaire items and operationalise aesthetics as a form of beauty [15, 20]. To control for intervening factors, most studies rely on participants viewing screenshots of user interfaces instead of interacting with technology.

Tractinsky et al. [49] described three processes as possible explanations of the relationship. The first two are derived from the judgment of human personality traits. First, Dion et al. [8] found that more desirable personality traits are attributed to physically attractive people than to unattractive people. They summarised their findings as “What is beautiful is good”, referring to the stereotypical association of physical beauty with other personality characteristics.

The second explanation concerns the *halo effect*, occurring when people infer unknown traits from a known trait of a person [47].

Since physical attractiveness can be easily judged, it affects the evaluation of other personal traits [8]. Thus, people who see attractive user interfaces infer that these would also be usable.

Third, Norman [31] explained the connection between aesthetics and usability with the user's mood, which is elevated by experiencing an aesthetic design. A positive emotional state of a person improves their creativity and flexibility in problem-solving [19]. A negative emotional state, on the other hand, promotes systematic and analytical problem-solving [40]. When problems arise while interacting with a website, a user in a good mood should be more likely to overcome problems creatively and therefore perceive these problems as less severe. A user in a negative mood will focus more on problematic details and judge usability less favourably [33].

Schrepp and Müller [38] propose the concept of clarity as a further explanatory variable. Clarity describes the perception of order, alignment, and visual complexity [39]. The authors propose that clarity is a common component of both, aesthetics and usability (for an earlier proposal see [26]). Questionnaires measuring visual aesthetics like the VisAWI [30] and questionnaires measuring usability like the User Experience Questionnaire [25] and the AttrakDiff [16] contain items about the visual clarity of the user interface. The original definition of clarity refers to a "clearly organised, simple, and thus easily grasped layout" [38]. With this, the authors circumscribe both factors of visual simplicity and perceptual fluency. As described earlier, the fluency effect entails that objects which are easy to perceive and process are also perceived to be more aesthetic [35]. Hassenzahl and Monk conjecture that classical aesthetics should be understood as a form of "visual usability" [17]. Moreover, it is reasonable to assume that user interfaces that are easy to perceive and process are also judged to be easy to use (perceived usability). It is on this conjecture that we grounded our study upon.

2.3 Research Questions and Hypotheses

Based on the theoretical background, two related research questions arise. First, we wanted to know whether the aesthetic-usability effect can be replicated using city websites as stimuli. Second, we wanted to find out whether manipulating processing fluency leads to websites being judged as more aesthetic and more usable. Moreover, we expect the aesthetic-usability effect to diminish when statistically controlling for processing fluency. Resulting from a pre-study, we selected city websites based on how fluent or disfluent experts and users rated them. Fluent and disfluent websites were then presented in a within-subjects design and participants rated their perceived aesthetics, perceived usability and subjective fluency.

We derived six hypotheses. First, we expected to replicate the aesthetic-usability effect based on previous studies that found a medium to a high correlation between aesthetics and usability (Table 1). Thus, we expected a positive correlation between judgements of aesthetics and usability (H1). Since we expected processing fluency to be the common factor underlying the aesthetic-usability effect, we expected fluency to be positively correlated with judgements of aesthetics (H2) and usability (H3). Importantly, statistically controlling for processing fluency should diminish the correlation between judgements of aesthetics and usability (H4). Further, when

websites were classified as fluent in the pre-study, we expected them to be judged as more aesthetically pleasing than websites that were classified as disfluent (H5). Similarly, when websites were classified as fluent in the pre-study, we expected them to be easier to use than websites classified as disfluent (H6).

3 METHODS

3.1 Pre-Study: Selection of Websites

The aim of the pre-study was to select fluent and disfluent stimuli for the main study. A panel of three fluency experts selected screenshots of city websites according to their expected processing fluency (for the selection process see Supplemental Material). As a manipulation check, these screenshots were then rated by 49 students. Processing fluency was measured with the item "How difficult or easy was it to study the website?", to which participants responded using a Visual Analogue Scale (VAS) ranging from 1 (*difficult*) to 101 (*easy*). This scale has been previously used in fluency research and is equivalent to a 5-item measure [11]. The result was a set of 50 websites which clearly distinguished fluent ($M = 70.84$, $SD = 10.01$) from disfluent websites ($M = 23.86$, $SD = 10.13$), $t(43) = 21.02$, $p < .001$. Figure 1 shows example images of a disfluent and a fluent website that were used in the main study.

3.2 Main Study: Sample and Procedure

Forty undergraduate students (different from those in the pre-study) participated in the study for course credit. Three were excluded (one because of too many errors in the memory test, two because of unexpected loading times $>10\text{sec}$ during the fluency blocks), so that a total of 37 participants were included in the analyses (30 female and 7 male, which is roughly representative of the gender distribution of our student body; average age 22 years). Students took part using their own computers; participation via a mobile device was excluded in advance due to the smaller screen size.

Due to COVID-19 restrictions, the study was conducted online. The study was designed and conducted with Gorilla Experiment Builder (www.gorilla.sc). The procedure was tested with eight pilot participants to check for length and comprehensibility. In the final version, a complete run required approximately 25 minutes.

First, the participants read the participation information (about voluntariness, anonymity, data protection) and gave informed consent. They were instructed to minimise possible distractions, to proceed without interruptions and to switch into full-screen mode. They then completed two practice trials, rating one fluent and one disfluent website that were different from the further websites presented in the study.

The main part was divided into three blocks, one for each dependent variable in the (fixed) order aesthetics – usability – fluency. Each block consisted of instructions followed by 20 stimuli (10 fluent, 10 disfluent), which were shown in randomised order (stimuli were the same in each block). In each trial, first, a black fixation cross was shown for 1000 ms [50] to direct attention to the centre where the stimulus appeared and to ensure that all participants started viewing from the same starting point [34]. Then, a stimulus was presented for 900 ms. This duration was deliberately short to prevent other website features such as semantic content from interfering with the assessment. After viewing a stimulus, participants



Figure 1: Example images of websites that were used in the main study (left: disfluent, right: fluent, both our own recreations).

made their judgement by clicking on the VAS and could proceed to the subsequent trial by clicking on a "Next" button. There was no time limit for giving an answer.

As an attention check, the main part was followed by a memory test with 18 stimuli. The first twelve stimuli were shown in randomised order, half of which had occurred in the main part (and the other half had not). Of these stimuli, one half were disfluent, the other fluent. Then, another six stimuli were shown in randomised order, all of which did not occur in the main part of the study (again one half disfluent, the other half fluent, as determined in the pre-study). The memory test was failed when more than two errors were made.

Towards the end, participants answered questions about their device use, gender, age, and field of study. They could comment on the study in a free-text field. Lastly, the participants created a code word with which they could request the deletion of their data if desired. The final screen thanked participants for their participation and explained the purpose of the study.

3.3 Study Design, Measures and Statistical Analysis

A within-subjects design with two levels was conducted. The independent variable was the fluency classification from the pre-study (fluent/disfluent). The dependent variables were perceived aesthetics and perceived usability. As a manipulation check we also measured processing fluency. With the aim of replicating previous studies (Table 1), and to keep overall mental workload low to not interfere with fluency measures, we used single-item measures (all scales ranged from 1 to 101):

- Perceived aesthetics: Although aesthetics has many dimensions [26], Jacobsen et al. [20] showed that beauty is the most prototypical aesthetic judgement. Thus, participants rated the item "Please give your impression of the website using the following word pair" on a Visual Analogue Scale (VAS) using the word pair *ugly* – *beautiful* [14, 50]. The wording corresponded to an item from the AttrakDiff subscale of attractiveness [16].
- Perceived usability: As a representative item of perceived usability, the expected ease of use was measured. Participants rated the item "I think the operation of the website is ..." on a VAS using the word pair *difficult* – *easy*. This form of

usability judgement is consistent with previous work on the aesthetic-usability effect [22].

- Processing fluency: Previous research has shown that a one-item measure of fluency is equivalent in validity to multi-item measures [11]. As in the pre-study, participants rated the item "How difficult or easy was it to study the website?" on a VAS with the verbal anchors *difficult* – *easy*. To better distinguish the fluency question from the usability question, a statement was added that "study the website" means "visually perceive and mentally process the website." This change was made during pilot testing.

To determine the correlation between aesthetics, fluency and usability, we computed a repeated measures correlation with the R package *rmcorr* version 0.4.2 [2, 3]. We checked the assumptions of linearity, normal distribution and homoscedasticity of residuals. Due to the heteroscedastic residuals, bootstrapping with 10,000 samples was performed to calculate confidence intervals [18]. To analyse the difference between disfluent and fluent websites, we calculated dependent samples t-tests. According to the Shapiro-Wilk test, the differences in fluency values were not normally distributed and contained one outlier. Bootstrapping with 10,000 samples was used to increase robustness and guard against the influence of this outlier [9, 18]. Because seven tests were conducted on one data set (including a fluency manipulation check), we used Bonferroni correction with an adjusted significance level of $\alpha = .0071$.

4 RESULTS

First, in line with the aesthetic-usability effect, we expected to see a positive correlation between perceived aesthetics and perceived usability (H1). The results indicate a large positive intrapersonal correlation between perceived aesthetics and perceived usability: $r_{rm_boot}(702) = .79$, 95% $CI_{boot} [.76, .81]$, $p < .001$.

Second, if processing fluency plays a role as a third variable behind the aesthetic-usability effect, we expected it to correlate highly with aesthetics (H2) and usability judgements (H3). Importantly, controlling for processing fluency should diminish the aesthetic-usability effect by a notable amount (H4). The results indicate a large positive intrapersonal correlation between perceived fluency and perceived aesthetics: $r_{rm_boot}(702) = .83$, 95% $CI_{boot} [.81, .85]$, $p < .001$. The results also indicate a large positive intrapersonal correlation between perceived fluency and perceived

Table 2: Comparison between fluent and disfluent websites including results of one-sided t-tests for paired samples.

	Fluent websites		Disfluent websites		<i>t</i> (36)	<i>p</i>	Cohen's <i>d</i>
	<i>M</i>	<i>SD</i>	<i>M</i>	<i>SD</i>			
Perceived aesthetics	75.61	8.94	23.16	11.65	21.14	< .001	3.47
Perceived usability	79.72	9.50	36.17	15.35	13.53	< .001	2.22
Perceived fluency (manip. check)	82.45	8.70	25.91	14.02	21.44	< .001	3.53

usability: $r_{rm_boot}(702) = .81$, 95% $CI_{boot} [.79, .84]$, $p < .001$. When controlling for the effect of processing fluency by calculating the partial correlation between aesthetics and usability, the original correlation of .79 reduces to .34, which is still statistically significant, $p < .001$.

Third, as we had manipulated the fluency of the websites, the distinction between fluent and disfluent websites was expected to also distinguish between aesthetic and unaesthetic (H5) as well as usable and unusable websites (H6). The data support these hypotheses: fluent stimuli were judged as more aesthetic and easier to use than disfluent stimuli (Table 2). The differences in both cases were highly significant, with a large effect size (Table 2). Finally, the manipulation check showed that disfluent websites were perceived as disfluent, and fluent websites were perceived as fluent. The difference was highly significant, with a large effect size (Table 2).

5 DISCUSSION

5.1 Summary and Interpretation of the Results

This work investigated whether processing fluency can explain the aesthetic-usability effect. All hypotheses were confirmed. The positive relation between perceived aesthetics and perceived usability (H1) is in line with previous research [21, 38], replicating the aesthetic-usability effect. This effect had already been found for different types of websites. Here, the correlation of .79 is higher than in most studies (cf. Table 1). Previous research suggests that both directions would be plausible to cause this effect, but current explanations favour a causal role of aesthetics on usability (instead of vice versa). Our data, however, show that another explanation is also likely: the correlation between aesthetics and usability could be caused by a third factor underlying both: processing fluency.

As predicted, positive correlations were found between perceived fluency and perceived aesthetics (H2) and between fluency and perceived usability (H3). Given that fluency was the manipulated variable, and fluency could distinguish aesthetic from unaesthetic websites (H5) and usable from unusable websites (H6), we assume that processing fluency caused the differences in perceived aesthetics and perceived usability. Previous research on the effects of processing fluency is rare and not conclusive: Tang et al. [45] found that the experienced ease of use is positively influenced by processing fluency. Chrobot [6], however, found no correlation between experienced usability and processing fluency. We provide new data on this and on the relationship between processing fluency and perceived aesthetics.

Finally, and most importantly, partialling out processing fluency greatly diminished the aesthetic-usability effect. The correlation between perceived aesthetics and perceived usability decreased

from .79 to .34; that is, the amount of common variance decreased from 62% to 12%. The partial correlation is still moderately high and significant, so other factors than fluency may still be at play. Fluency, however, could explain a large, and possibly the largest, chunk of the common variation. Again, our results support the view that there is a causal relationship between fluency on the one hand and aesthetics and usability on the other. Thus, previous findings showing an effect of aesthetics on usability as well as the reverse effect of usability on aesthetics [13, 51] could be explained by processing fluency as a third influencing factor. Future studies need to differentiate these effects as fluency is likely to be correlated more with classical than with expressive aesthetics [26], the relationship between fluency and aesthetics is probably not linear [37, 53], and other factors like context and task might play a role [5, 52].

5.2 Limitations

While the findings reported here have a potentially large impact on how we explain the aesthetic-usability effect, we also need to be aware of limitations that may impact the generalisability of our findings.

First, the selection process of fluent and disfluent websites in the pre-study may have exaggerated the differences between fluent and disfluent stimuli and homogenised the websites within the groups. All fluent websites, for example, contained a large banner image, while the disfluent websites contained much text in smaller font sizes. These characteristics could have directly influenced the judgements of aesthetics and usability, and, consequently, we would have overestimated the correlations between these variables and processing fluency. Future studies need to consider a greater range of stimuli across the whole fluency spectrum and opt for more heterogeneity. To enhance internal validity, it would be desirable to vary the fluency by systematically manipulating variables that are known to influence processing fluency (e.g., symmetry, contrast, font legibility, visual complexity). Furthermore, future research should look at other website categories than city websites [50] as well as other user interface types.

Second, in our study (as in previous studies, see Table 1) the participants judged screenshots of websites and did not interact with them. When browsing the web, user interaction with a website is more important for the user experience than only viewing [50]. Nevertheless, first impressions are formed while initially visiting a website by passive viewing, which is close to our screenshot assessment. By viewing screenshots rather than interacting with websites, participants would use less cognitive capacities [10] and thus there was less interference with measuring processing fluency. Still, to enhance the ecological validity of the findings, future studies need to go beyond viewing screenshots to include actual interactions

and performance measures of usability. The latter would also be desirable to avoid common-method bias.

Third, the participants were university students, and, in our case, the sample was biased towards young, computer-savvy and predominantly female participants. As this is not representative of the general website user population, further research should include broader samples.

6 CONCLUSIONS AND FUTURE WORK

The present work was the first to investigate the role of processing fluency in explaining the aesthetic-usability effect. A set of fluent and disfluent city websites was selected from a larger pool of websites in a pre-study to manipulate fluency. This set was shown to participants who rated these on perceived aesthetics, perceived usability and processing fluency. We not only found that perceived fluency, perceived aesthetics, and perceived usability were highly interrelated, but that controlling for processing fluency considerably diminished the aesthetic-usability effect. In addition, the fluency manipulation led to a distinction between aesthetic and unaesthetic as well as usable and unusable websites, thereby showing a causal relationship between fluency and aesthetics/usability. Although further research is necessary to address the limitations of this study, it seems that the repeatedly found aesthetic-usability effect can be explained by processing fluency as a third factor underlying both aesthetics and usability judgements. Instead of "what is beautiful is usable" [49], we should better say, "mind at ease makes beautiful and usable" (cf. [55]).

This change in perspective on what lies behind the aesthetic-usability effect has immediate practical implications. As it is not aesthetics that drives perceived usability, designers should not try to manipulate aesthetics to make things appear more usable. Rather, they should enhance the processing fluency of an interface to shape both aesthetics and usability. Fluency research has found several parameters that influence the processing fluency of a design, e.g., visual complexity, visual contrast or typographical legibility [1].

The field of fluency research offers a range of correlates of processing fluency that are relevant to HCI. Among these are credibility and trust, preference and liking, the perception of risk and customer loyalty [1, 12, 42]. Without fluency as the explaining variable, we might fall for other apparent but spurious correlations among these factors. We would propose, for example, an aesthetic-trust effect, a usability-risk effect, or a preference-loyalty effect, but behind all these, processing fluency would be the third explaining factor. While these effects are mainly speculative at present, it shows that HCI can possibly gain more than an explanation of the aesthetic-usability effect by intensifying the investigation of processing fluency in the future.

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